

SIBI KUMAR S

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PROFESSIONAL SUMMARY

AI & ML Developer with hands-on experience building and deploying computer vision (YOLOv8, OpenCV, Dlib), on-device LLM/conversational AI (Ollama, Mistral), and IoT-cloud pipelines (AWS IoT Core, MQTT). Proven track record delivering end-to-end intelligent systems, from a fully autonomous humanoid robot to production-deployed facial recognition and fleet-monitoring platforms. Strong foundation in embedded systems (Raspberry Pi, Arduino, ESP32), real-time data processing, and full-stack integration of AI models into working hardware products.

EXPERIENCE

AI Engineer — Flexidigit Technologies

Dec 2025 – Present

- Architected and deployed AWS IoT cloud infrastructure for real-time GPS tracking and fuel monitoring, enabling live fleet analytics across multiple vehicles and devices.
- Engineered real-time data communication pipelines using the MQTT protocol, connecting field hardware devices to AWS IoT Core for continuous, low-latency cloud ingestion.
- Led end-to-end hardware deployment for GPS (FMC650) and fuel sensor (Italon) devices, managing device provisioning, connectivity configuration, and cloud integration.
- Built scalable automation workflows supporting concurrent multi-device deployments, device shadow management, and cloud-side rule-based alerting for fleet operations.

EDUCATION

B.Tech — Artificial Intelligence & Data Science

May 2025

Dhanalakshmi Srinivasan University, Trichy

Class XII — Bio-Maths

March 2021

SVMHR Secondary School, Trichy

Class X

March 2019

Saividhyalaya C.B.S.E School, Trichy

PROJECTS

AI-Powered Humanoid Robot — Final Year Project

90 Days

Technologies: Python, Ollama (Mistral LLM), YOLOv8, Face Recognition, Speech Recognition, TTS

Hardware: Raspberry Pi, Arduino, Webcam, Microphone, Servo Motors

- Designed and built a fully functional humanoid robot capable of real-time voice interaction, face recognition, and object detection using YOLOv8.
- Integrated the Mistral LLM via Ollama for on-device natural language processing and conversational AI, eliminating dependency on cloud APIs.
- Implemented servo-motor-driven expressive hand gesture control synchronized with voice responses for natural human-robot interaction.

GPS Tracking & Fuel Monitoring for Logistics — Industry Project, Flexidigit Technologies

60 Days

Technologies: Python, AWS IoT, MQTT

Hardware: FMC650 GPS Device, Italon Fuel Sensor

- Designed a real-time data transmission pipeline from IoT field devices to AWS IoT Core to power live vehicle tracking and fuel analytics dashboards.
- Processed and normalized GPS coordinates and fuel sensor readings to drive fleet management reporting.
- Implemented a scalable MQTT communication layer supporting concurrent multi-device data streams to the cloud.

Quantum Vision X — IP Camera Monitoring System (Deployed Product)

30 Days

Technologies: Python, OpenCV, Dlib, MongoDB, Grayscale Processing

Hardware: Raspberry Pi 5, IP Cameras, NVR

- Built and deployed a network-based monitoring system that tracks real-time online/offline status of multiple IP cameras with centralized management.
- Implemented automated connectivity-failure alerts, reducing manual downtime detection and response time.

Quantum Vision — Facial Attendance System (Deployed Product)	40 Days
Technologies: Python, OpenCV, Dlib, MongoDB	
Hardware: Raspberry Pi 4, Pi Camera V2	
- Built and deployed a facial recognition attendance system with automated logging and repeated-entry alerts. - Integrated MongoDB for scalable attendance record storage and real-time retrieval.	
Ollama Voice Assistant — Individual Project	10 Days
Technologies: Python, Ollama, Text-to-Speech (TTS), Speech Recognition	
Hardware: Raspberry Pi, Microphone, Speaker	
- Developed a fully offline AI voice assistant on Raspberry Pi using the Ollama LLM for on-device natural language understanding. - Integrated speech recognition and TTS pipelines for seamless, low-latency voice input-output interaction.	
Attendance Management System with LLM — University Project (Team of 2)	50 Days
Technologies: Python, HTML, CSS, SQLite, LLM Integration	
- Built a web-based student attendance management system with an LLM-powered natural language interface for database queries. - Designed an intuitive frontend dashboard backed by SQLite for persistent, structured record storage.	
IoT Home Automation System — Individual Project	10 Days
Technologies: C, Arduino IDE, ESP32, Blynk IoT	
Hardware: ESP32 Microcontroller, Servo Motors	
Developed a Wi-Fi-enabled home automation system controllable via the Blynk IoT mobile platform.	

TECHNICAL SKILLS

Programming Languages: Python, C, HTML, CSS, SQL

AI / ML: YOLOv8, OpenCV, Dlib, Face Recognition, Ollama (Mistral LLM), NLP, Model Deployment, TTS, SpeechRecognition, Pandas

Cloud & IoT: AWS IoT Core, MQTT, Device Shadow, IoT Rule Engine, Blynk IoT, Real-Time Data Pipelines

Embedded & Hardware: Raspberry Pi, Arduino, ESP32, FMC650 GPS, Italon Fuel Sensor, Embedded Systems Programming

Databases: MongoDB, SQLite

Tools: Arduino IDE, VS Code, GitHub, Figma, Dialogflow, Pi Imager, MongoDB Compass

Core Competencies: Computer Vision, Conversational AI, Real-Time Systems, Human-Robot Interaction, Cross-Functional Collaboration, Technical Leadership

INTERNSHIPS

AI & ML Intern — Quantum Sharq (Offline, Stipend-Based)	90 Days
Worked on applied AI and ML projects in an offline, industry-aligned training environment.	
IoT Intern — One Stop.AI (Online)	45 Days
Completed hands-on IoT projects covering embedded systems programming and cloud connectivity.	

CERTIFICATIONS

Online: Data Visualization (IBM) | Data Analysis (IBM) | Artificial Intelligence (Great Learning) | Google Vertex AI Badge | Arduino vs Raspberry Pi (Great Learning) | AWS IoT Certification (Udemy)

Offline: Prompt Engineering (L&T Edu Tech)

PUBLICATIONS

"Design and Development of Attendance Management and Analysis System Using LLM" — Journal of Information Technology and Digital World (ISSN: 2582-418X), September 2024.

Available at: <https://doi.org/10.36548/jitdw.2024.3.007>

WORKSHOPS & CONFERENCES

- Conference on Innovation in Medical Technologies (ICIMT 2023) — Dhanalakshmi Srinivasan University, January 2023
- 2-Day Workshop on Intellectual Property Rights — Dhanalakshmi Srinivasan University, April 2024
- International Conference on Innovations and Applications of AI in Engineering & Agriculture (ICIAAIEA 2024), July 2024
- International Seminar on Industry 5.0 Skills — Dhanalakshmi Srinivasan University, July 2024

LANGUAGES

English — Professional Proficiency | Tamil — Native / Fluent | Telugu — Conversational